Birthday paradox
https://en.wikipedia.org/wiki/Birthday_problem
Statement: in a group of 23 people, at least 2 people share a birthday
Problem: given a group of N people what is the probability that at least 2 people share a birthday.

Consider the converse: possible groups where no people share a birthday.

365 choices * 364 choices * 363 choices * ...

=> probability that a group of N people all have distinct birthdays
= 1 * (364/365) * (363/365) * ...
= ∏ (1- k/365) k = 0 ... N

=> k = 20 p = 0.55631
k = 21 p = 0.5243
k = 22 p = 0.4927 (23 people)

Monty Hall
Factors: win/lose + change/no change
change + win = bad + change = 2/3 * 1/2 = 1/3
change + lose = good + change = 1/3 * 1/2 = 1/6
no change + win = good + no change = 1/3 * 1/2 = 1/6
no change + lose = bad + change = 2/3 * 1/2 = 1/3

Coin
Problem: Flip 2 coins. If one is heads, what is the probability that the other is tails?

H H no
H T yes
T H yes
T T

=> p = 2/3

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This
is
code
Monaco 14
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